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Three Ways to Miss a Grasp: A Same-Metric Comparison of Rule-Based, Learned, and Zero-Shot Vision-Language Prediction

Abstract

A robot cannot pick anything up until something decides where on the object to place the fingers and at what angle. That decision can come from hand-written rules, from a model trained to predict grasps from images, or from a general-purpose vision-language model never trained on grasping at all. All three were built and scored against the same external answer key, the Cornell Grasping Dataset's hand-labeled grasp rectangles, on one held-out split of 123 images covering 35 objects absent from training. Success was scored by the standard Cornell rectangle metric, an angle within 30 degrees of a labeled grasp and an intersection over union above 25%. The learned system (fine-tuned ResNet18) passed on 79.7% of test images, the rule-based system on 57.7%, and the zero-shot system (GPT-4o, one frozen prompt, averaged over five runs per image) on 12.4%. A second training round with a stable recipe and three seeds per architecture put the learned system at 84.0% (spread 0.5 points) and showed that the first round's ResNet18-over-ResNet34 gap was run-to-run noise. The rule-based figure is post-hoc rather than held-out, since a defect found after the test set was opened prompted a fix that raised it from 40.7%, and both numbers are reported throughout. The gap between the rule-based and learned systems is significant under McNemar's exact test, and the vision-language model trailed the rule-based system by 22.8 points even at its best-of-five accuracy of 35.0%, with an object-clustered 95% interval on that difference of [9.2, 35.6]. A secondary analysis split test images by whether their labeled grasps run diagonally. The three systems moved in three different directions, though on only 20 diagonal images, so that finding is exploratory. The model often named the correct part of the object while giving coordinates that landed in none of the labeled grasp rectangles, on 53.4% of calls against roughly 5% for the other

two systems, which suggested that GPT-4o might see where to grasp but fail to convert that into coordinates. A fourth run tested that directly. Shown twelve label-free candidate grasps drawn on the image and asked to pick one by number, the same model scored 18.7%, indistinguishable from choosing a mark at random (20.1%) on a menu whose ceiling was 79.7%, and it picked the same wrong kind of grasp, an end pinch along the object's long axis, in 61% of calls. Two further menus, four candidates at the centroid drawn at real size or at one uniform size, stayed at their random floors, and the long-axis preference persisted with every mark the same size. Removing the coordinate output does not recover performance, which points to the grasp choice itself, although the model's reading of the drawn marks was not separately measured.

Introduction

Before a gripper closes on anything, something has to choose a contact point, an approach angle, and how wide to open the fingers. Every piece of motion that follows depends on that choice being roughly right. A robot arm can be mechanically excellent and still fail constantly if the thing telling it where to grip is wrong, which makes grasp prediction one of the places where a small accuracy difference turns into a large behavioral difference. It is also a problem that has to be solved from thin information. In most real settings a robot gets a camera image and little else, with no model of the object, no label saying what it is, and no annotation marking the handle.

There are three broadly different ways to make that decision, and they are not variations on a single method. A programmer can write the rule directly, saying that mugs get grasped by the handle and bottles around the neck. A model can be trained on thousands of hand-labeled examples until it learns to map pixels to grasp rectangles. Or a general-purpose vision-language model, trained on web text and images and never shown a grasping dataset, can simply be asked in plain language where to grip. Each is a bet about where the useful intelligence in a robot's perception stack should live, whether in a person's explicit rules, in weights fitted to task-specific data, or in a large model's general knowledge of the world.

That third option is new enough that the field has not settled how to use it. Foundation mod-

els arrived in robotics quickly, and the literature is still working out which parts of a manipulation pipeline they should own. The question matters practically, because the three approaches demand very different resources. A lookup table needs only hand-specified rules. A learned predictor needs a task-specific annotated dataset and the compute to train on it. Prompting a hosted vision-language model needs an API key and almost no setup, which is exactly why it is tempting, and exactly why it deserves to be measured rather than assumed.

Comparisons across those three families are rarer than you might expect. Most published grasp-detection work benchmarks within a family, comparing one trained architecture against another on the same dataset. That produces useful leaderboards but does not tell you what you gain or give up by changing approach entirely. Rule-based baselines, when they appear, are often described rather than scored. Vision-language models usually appear embedded inside a larger system, where their individual contribution cannot be isolated. This project is a comparison where all three sit on the same test images, produce the same kind of output, and are scored by the same code.

The question is plain. Which approach actually works, and where does each one break down? The second half matters at least as much as the first. An accuracy number tells you a system is wrong. The shape of its errors tells you why, and that is the part that transfers to whatever somebody builds next.

The measuring apparatus is borrowed rather than invented, which matters more than it might seem. A grasp here is a rectangle with a center point, an orientation, and a width matched to how far the gripper opens, the representation Jiang et al. introduced with the Cornell Grasping Dataset (3304). Each image carries several labeled grasps rather than one, since most objects can be picked up in more than one valid way, and a prediction counts as correct if it matches any of them. The matching rule is that the predicted angle falls within 30 degrees of a labeled grasp and the intersection over union between the two rectangles exceeds 25%, the convention established by Lenz et al. (712). An external answer key was chosen deliberately. An earlier design would have had me write the ground truth myself and then grade my own rule-based system against it, which is close to circular. The dataset's existing annotations remove that problem and make these numbers

comparable to published ones.

The three systems are as follows. System A is the rule-based baseline, in which a COCO-pretrained detector identifies the object and a small fixed table maps the detected category to a grasp region, with a geometric fallback covering the images the detector cannot name. System B is the learned predictor, a small convolutional network built from scratch alongside fine-tuned ResNet18 and ResNet34 models, trained on the Cornell training split. System C is GPT-4o, prompted directly with the raw image and given no task-specific training whatsoever. System D is the same model shown a menu of candidate grasps drawn on the image and asked only to choose one, a follow-up run added after the first three systems were scored, to test the explanation their results suggested.

One thing this paper is not. It does not beat state-of-the-art grasp detection and does not try to. Published pixel-wise systems reach accuracies this work is not competing with. What this paper does is measure, under conditions held as close to identical as the design allows, how three families of technique fail, and argue that the shape of each failure carries information the accuracy number does not.

What it does establish can be stated plainly. The three systems ran on the same 123 held-out images through one shared scoring path. Their ordering survives the change of sampling unit from image to object and an object-clustered permutation test. The failure taxonomy then separates the three families along lines that a single accuracy number does not show, and the fourth run rules out the most natural explanation of the vision-language model’s failure.

Methods and discussion

Dataset and evaluation

The dataset used for evaluation is the Cornell Grasping Dataset, roughly a thousand images of household objects, each with hand-labeled grasp rectangles (Jiang et al.). A prediction counts as correct when its angle falls within 30 degrees of some labeled grasp and its rectangle intersection over union with that grasp exceeds 25%, the convention established by Lenz et al. Cornell’s annotations are not exhaustive, so passing this metric means matching a labeled grasp, not that the grasp

would physically succeed. The Jacquard dataset was considered as an alternative, but its full distribution requires a signed agreement and an approved non-free-email request. Object identity was the hardest problem in building the split. No object may appear in both training and test, or a model can memorize its grasp rectangles from one split and be credited for recognizing the same object in the other. Cornell provides no object-identity labels, so identity had to be reconstructed from the image sequence. Comparing raw pixel differences between consecutive frames failed, because the dataset deliberately rotates each object between shots and same-object and different-object scores overlapped. Instead, each frame was segmented against the photography platform, and the resulting object regions were compared to decide whether two frames showed the same object or different ones.

Acceptance thresholds were set asymmetrically, since merging two different objects into one group is less harmful than splitting one real object into two. The “different-object” threshold was set below the lowest score seen for any confirmed same-object pair, while the “same-object” threshold stayed loose. A 30-boundary stratified sample was classified by both an AI assistant and a human, blind to each other’s calls. The AI’s classifications matched the human’s on 86.7% of its confident calls but only 40.0% of the calls it flagged as uncertain, so every uncertain call and every confident “different” call was reviewed by hand. Confident “same” calls were accepted unreviewed, since an error in that direction is harmless. The result is 234 objects across 883 images, with 620 for training, 140 for validation, and 123 for test, as shown in Table 1. Published descriptions of Cornell normally report around 240 objects, and the shortfall here is expected rather than surprising, since the thresholds were deliberately biased toward merging. Every merge that should have been a split lowers the object count while keeping the merged group on one side of the division, which is the error direction this design accepts on purpose. Each object has at least one grasp rectangle, and most have several, since most objects can be picked up more than one way.

System A, rule-based baseline

System A is better described as a detector-assisted heuristic baseline than a purely rule-based one. It detects the object with a COCO-pretrained detector, which is itself a learned component,

then looks up the detected category in a fixed table of grasps. Nothing in it is learned from grasp labels, which is the sense in which it is rule-based. The table was committed before any evaluation code existed, so no entry could be tuned toward a score. Since COCO’s categories miss more than half of what Cornell photographs, a geometric fallback that does not rely on object recognition covers the objects the detector misses, and that fallback handles the majority of test images.

One qualification belongs here rather than in Results. System A was scored once at 40.7% before the detector was found to be boxing background clutter. The guard that fixed it was added after those test predictions had been inspected, so although the recalibrated constant used training data only, the decision to add the guard was prompted by test behavior. That makes 57.7% a post-hoc corrected figure rather than a clean held-out one, and it weakens the comparisons that rest on it, including the McNemar test against System B and the interval comparison against System C. Both numbers are reported everywhere the result appears.

A second limitation follows from the table’s design. COCO’s 80 categories do not cover most of what Cornell photographs, so many objects have no table entry to begin with. And regardless of path, System A’s rectangle orientation is limited to 0 or 90 degrees, the only two orientations the table and the box-based rule can express, so any grasp that requires an angle in between is one System A cannot produce.

System B, learned grasp predictor

System B regresses a grasp rectangle directly from an image. A backbone network extracts features, pools them, and a regression head predicts the rectangle’s parameters. Orientation is encoded as the sine and cosine of twice the angle rather than a raw angle between 0 and 180 degrees. This accounts for the fact that a grasp rectangle rotated 180 degrees describes the same physical grasp. A raw-angle representation would score that rotation as an entirely different grasp, while the sine and cosine encoding treats it as identical. Redmon and Angelova (1316) and Morrison et al. both used this same encoding.

System B trained three architectures on the same training split, a small network built from scratch, a fine-tuned ResNet18, and a fine-tuned ResNet34. The loss was computed against every

labeled grasp on an image, with only the best match backpropagated. Averaging toward all of an image's grasps would pull a handle grasp and a rim grasp toward a point on the object where neither is valid. Augmentation applied the same affine transform to the grasp rectangle's corners as to the image's pixels, rather than hand-writing a rule for how the angle behaves under a flip.

System B was trained twice. The first round trained each architecture once, at one seed, with a constant learning rate, and kept whichever of roughly a hundred epochs scored highest on validation. Validation accuracy on 140 images swings seven to twelve points between consecutive epochs, so that maximum is a lottery and a single run cannot separate one architecture from another. The second round kept the model, loss, data, metric, and split, and changed only the recipe: a cosine learning-rate decay over a fixed 100 epochs, an exponential moving average of the weights, three seeds per architecture, and no checkpoint selection at all, since the reported model is the final-epoch averaged weights. Test-time augmentation, eight dihedral views of each image with the prediction carried back through the same corner transform used in training and the medoid rectangle kept, was added and its use decided on validation. Four rules were written down before the second round touched the test split: the architecture is the one with the higher mean final-weight validation accuracy across seeds, augmentation is used only if it raises that mean, the headline is the mean over seeds of the final weights, and every configuration that touches test is reported. A 60-epoch budget was tried first and rejected on validation alone, since both architectures were still improving when the decay ended. Opening the test split a second time is a departure from the one-opening plan and is reported as one.

System B's accuracy can look low next to two published Cornell results, but neither is an even comparison. GR-ConvNet reports 97.7% image-wise and 96.6% object-wise on an extended 1,035-image version of Cornell (Kumra et al. 9626), and only the object-wise figure is on the same footing as anything here. Redmon and Angelova report 84.9% object-wise (1316). GR-ConvNet is pixel-wise. It predicts a grasp at every pixel in a single pass. System B does nothing of the kind. It pools a backbone's features and regresses one rectangle for the whole image, the same design Redmon and Angelova published, whose 84.9% object-wise score is the fairer target, as laid out

in Table 2. Every remaining gap points the same direction. Redmon and Angelova used RGB-D and roughly 3,000 augmented examples per image, while System B used RGB only and 620 real training images.

System C, zero-shot vision-language model

System C prompts GPT-4o with each test image and asks where the two gripper fingertips should touch, with no training or fine-tuning on grasping. The prompt was developed on a batch of 30 training images and frozen before test was opened.

The configuration was as follows. The model was called as `openai/gpt-4o` through the OpenRouter API, at a 400-token cap, with temperature left unset so the provider default applied and no seed pinned, since run-to-run variation was one of the things being measured. Images were sent at the dataset’s native 640 by 480 resolution with no resizing, and the prompt told the model that coordinates were in pixels of that image. Each repeat was an independent single-turn completion with no prior messages, so the five repeats of an image are five genuine draws rather than one conversation. Transport failures were retried up to three times, but a reply that arrived and then failed to parse was never retried, since re-rolling malformed content would quietly convert a single-shot baseline into a best-of-N one.

System C was not asked to output an angle. It returned two fingertip points and a jaw width as JSON, and those three values determine all four rectangle corners, which were then passed through the same frozen function that produced every ground-truth angle in the dataset. No rectangle parameter was supplied by me, and no radians-versus-degrees convention had to be matched by hand. Each image was prompted five times, and all five results are reported.

System C’s 12.4% does not mean vision-language models are bad at spatial tasks. In several published grasping systems built on the same kind of model, the model’s output is a choice, a label, an order of operations, or an image, and the pose of the gripper comes from something else. FreeGrasp asks the model to decide which object to grasp and in what order, using marks placed on the image (Jiao et al.). Lan-grasp asks the model which part to grasp, then hands the pose to a conventional planner (Mirjalili et al.). ThinkGrasp uses the model to plan clutter removal (Qian et

al.). Set-of-Mark asks the model to select one of several labeled regions (Yang et al.). VLAD-Grasp asks the model to draw the grasp using a virtual gripper (Kulshrestha et al.).

In none of them does the model emit the pose coordinates itself, which is exactly what System C is built to test. Does the gap come from the coordinate-emission step, or from the model’s inability to see the object?

Of 615 calls, 365 failed on both angle and overlap simultaneously, far more than either failure alone. A supplementary check asked whether each predicted center fell inside the convex hull of an image’s labeled grasp rectangles, the loosest spatial test a prediction can fail. System C failed it on 53.4% of calls, against 7.0% for System A and 6.5% for System B. That gap is real, though the comparison isn’t quite even, since System B was trained on this distribution and System A’s center comes from a detected box or a segmentation, so both are close to guaranteed to land on the object. System C is the only one free to place a center anywhere. Reading the model’s written explanations alongside its coordinates on a sample of failures, the explanation frequently named a real part of the object while the coordinates in the same reply landed somewhere unrelated. That reading was informal rather than a blinded coding study with a fixed rubric, so it should be treated as the observation that motivated the hypothesis, not as a measurement of how often the pattern holds.

Wang et al. document the same disconnect in an unrelated domain, with models describing a correct solution to a visual puzzle in text, then clicking hundreds of pixels away from it (Wang et al., sec. 5.3.2). The tasks differ, but the failure, text and coordinates disagreeing within one response, is the same.

System D, marked-candidate selection

System D is the test that the coordinate-binding reading calls for. If GPT-4o can see where to grasp but cannot express it as pixel coordinates, then removing the coordinate output should recover its choice. So the same model was shown the same test images with twelve candidate grasps drawn on a zoomed crop of the object and asked to return the number of the one it would use. It never emits a coordinate.

The candidates are generated without reading any grasp label. The platform segmentation already used for the object-wise split gives the object’s mask; principal component analysis of that mask gives its centroid, long axis, and extent. Candidates are three positions along the long axis (the centroid and 35% of the extent to either side) crossed with four jaw-travel directions 45 degrees apart, starting from the short axis, so every possible grasp angle lies within 22.5 degrees of some candidate, inside the metric’s 30 degree tolerance. Opening and jaw width use System A’s train-calibrated constants. Each candidate is drawn as two thick fingertip pads joined by a thin line with inward arrowheads, in its own color, with a numbered badge. The numbering is re-shuffled on every call, so a preference for a number cannot pass as a preference for a grasp, and self-agreement is measured on the physical candidate rather than on its label. Eleven test images have no segmentation mask and so no candidates; they count as misses for System D and for every reference row it is compared with, so the comparisons stay paired.

Because the menu is fixed before the model sees it, the run comes with its own floor and ceiling computed on the same candidates: the expected accuracy of a uniformly random choice, and whether any candidate passes at all. It also yields a fourth non-learned baseline for free, the candidate at the centroid closing across the short axis, which unlike System A can express a diagonal grasp.

The prompt was developed on the same 30 training images System C used, two repeats each, and judged on parse rate and on whether the model was reading the glyph as intended. The first version drew each candidate as a thin line with small end ticks. The model chose the candidate running along the object’s long axis in 34 of 58 calls while describing a grasp “across the narrowest part”, which could have meant it was reading the line as the finger plates rather than the closing direction. The second version drew explicit pads and arrowheads and described them in the prompt. The behavior did not change (41 of 58), which is what licenses reading the test result as the model’s preference rather than a drawing artifact. Accuracy on those development images was at the random floor under both versions and was not used to choose between them. The second version was frozen, and the test split was called once, five times per image.

One confound remained after that run. On thin objects the correct marks are drawn with the small opening the labels themselves use, so they are the smallest marks on the image, and a preference for large marks would mimic a preference for the long-axis grasp. Two further runs close it. Both use four candidates at the centroid only, the short-axis direction and three rotations of it, so position is removed and the end-straddling candidates do not exist. In the first, each mark is drawn at its real opening. In the second, the same four candidates with the same scored geometry are drawn at one uniform length, and the prompt gains a single sentence saying the length is not to scale. Each was parse-checked once on the 30 training images and then called on the test split five times per image, under its own sentinel.

The orientation axis

Each system’s design predicts something different about how orientation should affect it. System A’s representation cannot express a diagonal grasp, System B’s encoding handles rotation by construction, and System C has no structural position on the question at all.

An image counts as axis-aligned when any one of its labeled grasps sits within 15 degrees of 0 or 90 degrees, and as diagonal only when none does. The rule is deliberately conservative, since a system that can emit only axis-aligned rectangles still has a valid target whenever a single labeled grasp is axis-aligned. It is also looser than the 30 degree scoring tolerance, so a diagonal image whose labeled grasps sit 20 degrees off axis is still reachable by an axis-aligned prediction. That is why System A scores 40.0% on the diagonal stratum rather than near zero. Splitting the test set that way produced exactly the predicted pattern. System A dropped 21.2 points on the 20 diagonal images relative to axis-aligned ones, System B gained 12.3 points, and System C moved 2.0 points, as Fig. 5 shows.

An axis-aligned box being unable to represent a diagonal grasp follows from the representation alone, and a rotation-invariant encoding handling rotation is the entire reason that encoding exists, so neither point here is a discovery on its own. What I could not find precedent for was using this split as a diagnostic. It checks whether each system’s accuracy rises or falls along an axis defined by the dataset’s own grasp-angle annotations, rather than reporting a single train and test

generalization number.

The diagonal stratum holds only 20 images, and the 95% intervals around System B’s and System C’s per-stratum accuracies are wide enough that their point changes there are not reliably distinguishable from noise. Only System A’s drop was predicted from its representation before the stratum was scored, which makes it pre-registered rather than found afterward. Pre-registering a direction is not the same as confirming an effect, and no interaction test between system and stratum was run, so this remains exploratory too. System C’s near-flat response is at least consistent with the coordinate-binding hypothesis explored later, since a system with no working orientation signal should look flat on this split, and it does.

A second split, by the number of grasps labeled per image, was tested as a difficulty proxy. More labeled grasps should mean more chances for a prediction to match. It did not behave that way. All three systems did worst on the images with the most labeled grasps. The likely explanation is that annotators labeled more grasps on objects that afford more of them, so the count tracks object complexity rather than how generous the metric is.

What this means

This explanation makes a testable prediction. Removing System C’s ability to output coordinates directly should help if the problem is binding reasoning to those coordinates rather than perceiving the object, and should make no difference if the problem is perception. System D ran that test, and the prediction failed. Given a menu of twelve candidates with a 79.7% ceiling, GPT-4o chose a passing grasp on 18.7% of calls, against 20.1% for a uniformly random choice from the same menu, a paired object-clustered difference of -1.4 points $[-6.6, +4.0]$. Its gain over its own coordinate output, 6.3 points $[-1.6, +15.0]$, does not exclude zero either. Two published results had seemed consistent with the binding reading, though neither was designed as a test of it, and they are worth keeping in view because they show what a menu-based design can do when the menu and the model are matched. Set-of-Mark lets zero-shot GPT-4V beat a fine-tuned specialist on RefCOCOg purely by replacing coordinate output with region selection (Yang et al.). VLAD-Grasp scores far above System C on this same dataset, training-free, by having the model draw the grasp instead of

stating it (Kulshrestha et al.).

VLAD-Grasp is the comparison that most complicates System C’s result, but two things keep it from being a number that can be set directly against 12.4%. It does not simply ask for a picture instead of a coordinate. It then predicts depth, segments the object, and aligns 3D point clouds to recover a pose, a pipeline System C has no equivalent to. It also counts a grasp correct on overlap alone, with no angle requirement, and reports 91.4% under that looser test on a set of 70 unseen Cornell objects. That the same evaluation scores GR-ConvNet at 72.1%, against the 97.7% GR-ConvNet reports for itself, shows the two protocols are not directly comparable. The dropped angle criterion is what matters here, since orientation is the variable this paper’s results depend on most. It is also a different model rather than the same one, since the current revision of that paper names GPT-5 as its primary model, so it evidences what a vision-language model can do without emitting coordinates, not what GPT-4o specifically can do.

Even with that caveat, the contrast between VLAD-Grasp and System D is informative in a way System C alone was not. Both remove the coordinate output. One works and one does not, and the one that works also adds depth, segmentation, and a geometric pose recovery, so the work in that pipeline is being done by the geometry, not by the model’s choice. System D’s failure is specific. In 61% of calls it chose the one candidate whose jaws travel along the object’s long axis near one end (70% chose that orientation at some position), a type of grasp that passes the metric 17% of the time, while the centroid candidate closing across the short axis, which passes 61% of the time, was chosen in 6%. Its reasoning for the preferred choice was typically that it “grips the narrow top edge near the center of mass”. From a top-down photograph, two pads straddling the end of an elongated object do look like they pinch a narrow part. In three dimensions the second finger would land on top of the object rather than beside it. The model’s stated confidence was “high” on 556 of 560 calls, of which 21% were correct. So the failure that the center-outside-hull diagnostic exposed in System C is better described as a failure of grasp reasoning from a single 2-D view than as a failure to bind text to coordinates. A rendering explanation was tested and ruled out. With four candidates at the centroid and every mark drawn at the same size, the model still chose the long-axis direction

in 45% of calls against a 25% chance rate, and chose the short-axis candidate, the one that passes on 55% of images, in 17%, below chance. Its accuracy sat at the random floor on all three menus. The preference is weaker without end-straddling candidates (70% on the full menu, 41 to 45% on the centroid-only menus), so part of the full-menu result was the end-pinch illusion specifically and part is a plain preference for closing along the long axis.

The sine and cosine encoding dates to 2015, learned models have beaten rule-based baselines on Cornell for a decade, and imprecise vision-language coordinates are already documented elsewhere, so none of this is new by itself. Putting a number on that last weakness, on a standard benchmark against two baselines, and then testing the most natural account of it and finding it wrong, is what this experiment adds. A side result came free with the menu: a rule that closes across the segmented object's short axis at its centroid, with no learning and no detector, scores 55.3%, statistically level with System A's post-hoc 57.7% (-2.4 points $[-10.1, +5.4]$), and it is the first label-free baseline in this study that can express a diagonal grasp.

One correction to the first draft of this paper came from re-running its own experiment. The first round's nine-point gap between ResNet18 and ResNet34 was explained there as the smaller network suiting a small dataset. Three seeds per architecture under a stable recipe put them within a point of each other. A single training run on 620 images, with a checkpoint chosen from a hundred noisy validation epochs, is not evidence about architecture, and the explanation that fit it was invented after the fact.

A one-million-parameter network built from scratch never learned the task by any reasonable standard, reaching 23.6% after a learning-rate sweep confirmed that wasn't a tuning artifact. It still beat GPT-4o (System C) by 11 points, which I did not expect.

Three bugs in my own code were caught before any result was reported, each of which would have produced a plausible but wrong number. All three turned up the same way. An automated result didn't match an independent calculation, and neither was trusted until the disagreement was explained.

Limitations

This test was limited to using a single split of the dataset relative to the five-fold cross-validation that is used by most published results using the Cornell dataset. Seven of the 35 object groups in the test split contain only a single image of the object, so the results for those objects are not as precise as they could be. The test split was opened twice for System B, once per round, and the second round’s selection rules were fixed beforehand so that the second opening could not steer any choice, but two openings are still two. The round-1 numbers rest on one seed each; the round-2 numbers on three. Jacquard, considered as a second dataset source, was unavailable during this study, so no results were performed to test the generalization of these techniques across datasets. An early formatting check on roughly 550 grasp rectangles likely touched a small number of images that later landed in the test split. No threshold, constant, or lookup entry was derived from that check, but it should be disclosed.

The three systems did not receive comparable engineering effort, which cuts against reading this as a controlled comparison. System B got three architectures, a five-point learning-rate sweep, a tuned loss weight, and validation-based model selection. System C got one prompt, frozen after development on 30 training images and never revised. Freezing it protected test-set integrity and I would do it again, but prompt design is the vision-language equivalent of architecture search, and I ran architecture search for one system and not the other. So 12.4% measures what one un-iterated prompt achieves, not a ceiling for zero-shot GPT-4o. The gap to System B is wide enough that prompt iteration is unlikely to reverse the ordering, but the specific number should not be quoted as the model’s best. System D shares this limit and adds one of its own: three menus from one candidate generator, all drawn in one glyph style on a zoomed crop. A different mark style, or marks placed by a segmentation the model itself produced, was not tested.

System C’s potential contamination by the dataset was examined. While the dataset is a publicly available dataset that is mirrored by many organizations, a test of ten images of objects of various categories asked the language model to identify the source of the images returned no recognitions. This is a weak indicator of contamination of the model by the dataset, but contamination

of the language model would only have led to an increase in System C's score, which is already the lowest of the three systems. So the failure of System C in comparison to the other models holds even in the worst-case scenario of contamination of the model by the dataset.

Every interval and significance test reported here treats its unit of analysis as independent, and none of them are. System C's interval is computed over 615 calls, but those are five repeats of 123 images, not 615 independent trials, so it is narrower than the data support. The image-level intervals and McNemar tests for Systems A and B have the same problem one level up, since the 123 test images are grouped within 35 objects and images of one object are not independent of each other. The stated population is unseen objects, so the correct sampling unit is the object. The object-clustered intervals and differences reported in the results section are the primary ones, and every Wilson interval quoted per system is image-level and narrower than the data support. Weighting each object equally rather than by how many images it contributes lowers every system, System A to 51.3%, System B to 75.4%, and System C to 10.5%, because test object groups range from one image to nine. The ordering is unchanged.

The findings of the results relative to the orientation of the objects was based on a sample size of only 20 objects. Additionally, the other findings of this test were made of one model, one method of posing the question to that model, and the one aspect of the task of grasping that is examined, the coordinates of the grasp, so these results describe that one setup rather than vision-language models generally.

All of the inputs to the models used in this test were images in RGB format only. No depth information from the objects was used as an input to any of the tested models.

Results

The data split

Table 1. Object-wise split of the Cornell Grasping Dataset. No object appears in more than one split.

Split	Objects	Images
Train	164	620
Validation	35	140
Test	35	123
Total	234	883

The split covers 234 objects across 883 images, divided roughly 70/15/15 by object count. No object appears in more than one split.

Of the 884 frame boundaries examined, 251 (28.4%) fell into the ambiguous band. 139 of those were decided by hand, and the remaining 112 were confident same-object calls accepted without review. Six stayed genuinely uncertain after full review. Four carried a lean toward “same object” and were resolved that way, and the other two had no lean. One image from the smaller side of each was dropped instead, accounting for the 883 images retained out of 885.

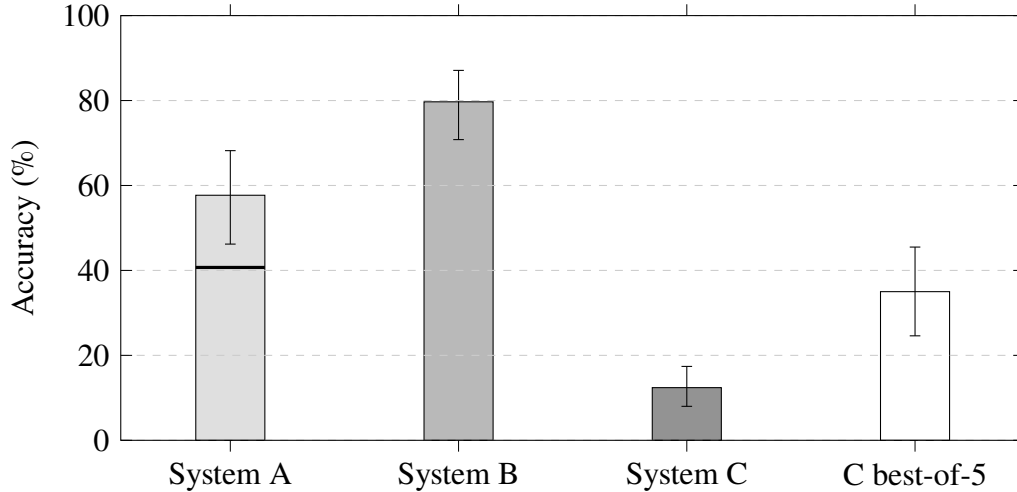


Fig. 1. Test-set accuracy with 95% object-clustered intervals, the primary intervals in this paper. System A 57.7 [46.2, 68.2], System B 79.7 [70.8, 87.1], System C 12.4 [8.0, 17.4], and System C’s best-of-five ceiling 35.0 [24.6, 45.5], which requires five calls per image rather than one. The horizontal rule across the System A bar marks 40.7%, the figure that system scored before the post-hoc geometric guard was added, so the size of that correction is visible here rather than only in the text. System B’s bar is the round-1 single-seed figure; the three-seed round-2 figure is 84.0 [73.9, 92.1], reported in Table 3.

The ordering in Fig. 1 is the paper’s main result, and the per-system subsections below give the counts behind each bar.

System A

System A achieved 57.7% on the test split (71 of 123), with a 95% Wilson interval of [48.9, 66.1]. Of the 52 failed predictions, 13 missed the orientation step alone.

The two paths performed differently. The detector fired on 39.8% of test images (49 of 123). Counting every image where it fired on nothing as a failure gives 22.8% (28 of 123), while accuracy on the images where it did fire was 57.1% (28 of 49). The fallback geometric system attempted 66 images and found no grasp on 8. Accuracy by object category was uneven, with all apples correct (6 of 6) but only 8 of 11 cell phones, none of the cups or bowls, and none of the laptops.

The table was evaluated once at 40.7% before the detector was found to be boxing background clutter. A geometric guard requiring the box to contain the object’s centroid, plus one constant recalibrated on training, raised the result to 57.7%. Both figures stay on record. The

detector-only number moved only from 22.0% to 22.8%.

System B

System B is compared with its closest published counterpart in Table 2.

Table 2. System B compared with Redmon and Angelova’s Direct Regression, the closest published architecture.

	Redmon and Angelova (2015)	System B
Architecture	Global regression	Global regression
Orientation encoding	sin/cos of twice the angle	sin/cos of twice the angle
Input	RGB-D	RGB only
Training data	~3,000 augmented per image	620 real images
Backbone	AlexNet	ResNet18, then ResNet34
Object-wise accuracy	84.9%	79.7% (round 1), 84.0% (round 2)

It achieved 79.7% on the test split (98 of 123), with a 95% Wilson interval of [71.7, 85.8]. Its mean angle error of 3.9 degrees is the average, over all 123 test images, of the smallest angle difference between the prediction and any labeled grasp on that image, regardless of whether the prediction overlapped that grasp or passed the metric. Restricted to the 98 images it got right and measured against the labeled grasp the metric actually matched, the mean angle error is 7.7 degrees. The first figure says the network’s orientation is usually close to some valid orientation on the image; the second says how precisely it orients the grasp it actually places. System B’s ResNet34 model achieved 70.7% (87 of 123) with an average intersection over union of 0.424 of its grasps with the labeled grasps. The from-scratch model achieved 23.6% (29 of 123) with an average angle error of 24.2 degrees. Parameter counts were 11.3M, 21.4M, and 1.0M respectively.

In round 1, System B’s ResNet18 outperformed ResNet34 on validation as well as test (83.6% versus 77.1%), so the selection held across both splits; the second round below shows that gap was noise. The validation-to-test gap (3.9 points) sits well inside the 6.7 to 11.9 point swings seen between epochs during training, so it isn’t evidence of overfitting. The from-scratch

network got its own five-point learning-rate sweep on validation, scoring 21.4% at $3e-5$, 27.1% at $1e-4$, 26.4% at $3e-4$, 20.0% at $1e-3$, and 22.9% at $3e-3$. The rate of $1e-4$ was selected.

System B's ResNet18 model missed 25 of 123 test images. Of those missed images, 4 failed on determining the angle of the object to grasp, while 15 failed on the overlap of the bounding box around the object with any labeled grasp in the image. ResNet34 failed to detect grasps in 36 images, while the from-scratch model missed 94 images. One of those angle-only failures is shown in Fig. 2.

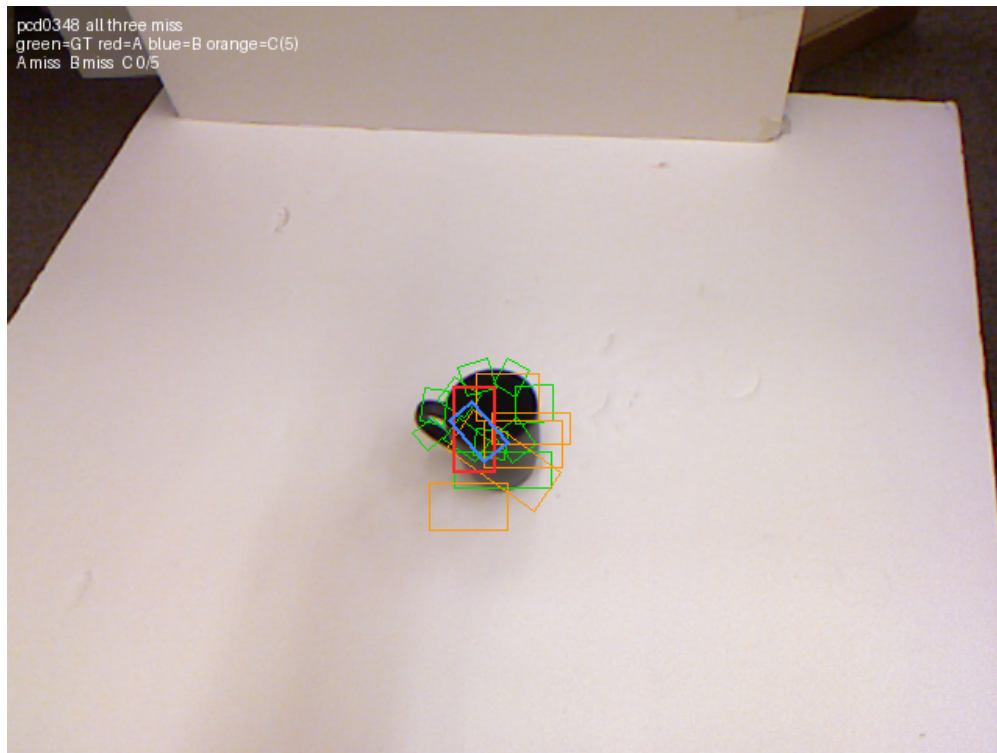


Fig. 2. An orientation-only failure. The predicted grasping rectangle correctly overlaps the object (intersection over union of 0.38) but rotates 47 degrees from any labeled grasp. Of the four ResNet18 angle-only failures (pcd0676, pcd0348, pcd0824, pcd0316), only pcd0348 has a rendered sheet, which is why it was chosen.

The second round, in Table 3, chose ResNet34 by a margin of 0.2 validation points (81.4% against 81.2% mean final weights over three seeds) and used augmentation, which lifted mean validation from 81.4% to 83.3%. On test the three ResNet34 seeds scored 84.6, 83.7, and 83.7%, a mean of 84.0% with a standard deviation of 0.5. ResNet18 under the same recipe scored 83.7, 80.5, and 83.7%, a mean of 82.6%. Re-running the first-round recipe at two new seeds gave validation-

selected checkpoints scoring 82.9% and 80.5% on test, so the first round’s 79.7% was a low-side sample from a recipe whose seed spread is about 1.6 points at the checkpoint and much larger at the final weights, where the same two runs scored 80.5% and 62.6%. Averaged final weights beat the validation-selected checkpoint of their own run on five of six second-round runs. Test-time augmentation added between 0.0 and 5.7 points across the six runs, about three on average.

Table 3. System B, both rounds, on the same 123 test images. Round 2 numbers are means over three seeds of the final-epoch averaged weights; no checkpoint was selected.

Round	Configuration	Test accuracy
1	ResNet18, seed 42, best validation checkpoint	79.7%
1	ResNet34, seed 42, best validation checkpoint	70.7%
1 recipe	ResNet18, seeds 0 and 1, best validation checkpoint	82.9%, 80.5%
2	ResNet18, three seeds, final weights, with augmentation	82.6% (83.7, 80.5, 83.7)
2	ResNet34, three seeds, final weights, with augmentation	84.0% (84.6, 83.7, 83.7)

Object-clustered, round 2’s 84.0% carries an interval of [73.9, 92.1]. It leads System A by 26.3 points [14.8, 38.3], permutation $p = 0.0002$, and System C by 71.7 points [61.3, 80.9]. Its 4.3-point lead over round 1 has an interval of $[-5.8, +13.7]$ and does not exclude zero, so the second round is best read as the same result with its uncertainty measured rather than as an improvement. Round 1’s finding that ResNet18 beat ResNet34 by nine points, which the first draft of this paper explained by the smaller network suiting 620 images, does not survive: the two are level.

System C

System C achieved 12.4% accuracy on test images (76 of 615 calls), with a 95% Wilson interval of [10.0, 15.2]. Each of the five independent runs achieved 13.0%, 14.6%, 13.8%, 10.6%, and 9.8% accuracy. The best-of-five attempts achieved 35.0% (43 of 123). A consensus rule that scores whichever single repeat agrees most closely with the other four, since continuous rectangles cannot be majority-voted directly, reached 12.2%. Neither of these is the headline figure. Both require either multiple calls per image or knowing the answer in advance. The model returned a

parseable reply on 614 of 615 calls (99.8%), and every non-parsing call counts as a miss in the headline figure.

Table 4. Which criterion each prediction missed, computed through one shared scoring path for all three systems.

Outcome	System A (of 123)	System B (of 123)	System C (of 615)
Correct	71 (57.7%)	98 (79.7%)	76 (12.4%)
Angle only	13 (10.6%)	4 (3.3%)	85 (13.8%)
Overlap only	18 (14.6%)	15 (12.2%)	88 (14.3%)
Both	13 (10.6%)	6 (4.9%)	365 (59.3%)
No prediction	8 (6.5%)	0 (0.0%)	1 (0.2%)

The failure taxonomy in Table 4 shows where the three systems differ most. For the geometric check described in Methods, System C’s predicted center fell outside the convex hull containing every labeled grasp on 53.4% of calls (328 of 614). System A showed failures of this type on 7.0% of the calls in which it attempted a grasp (8 of 115 scored calls), and System B on 6.5% (8 of 123).

Table 5 puts the three side by side.

Table 5. Predicted grasp centers falling outside the convex hull of an image’s labeled grasps. This is a diagnostic across scored calls, not a partition, so it overlaps the failure categories in Table 4 rather than adding to them.

System	Calls scored	Center outside hull
System A	115	8 (7.0%)
System B	123	8 (6.5%)
System C	614	328 (53.4%)

Agreement across System C’s five runs was low. Mean self-agreement was 22.0%, with a mean pairwise angle spread of 5.8 degrees and mean pairwise intersection over union of 0.13. The distribution of how many of five repeats scored correct per image appears in Fig. 3. That puts 81 of

123 images (65.9%) at a fully consistent outcome, leaving 42 where the same model on the same pixels sometimes passed and sometimes didn't.

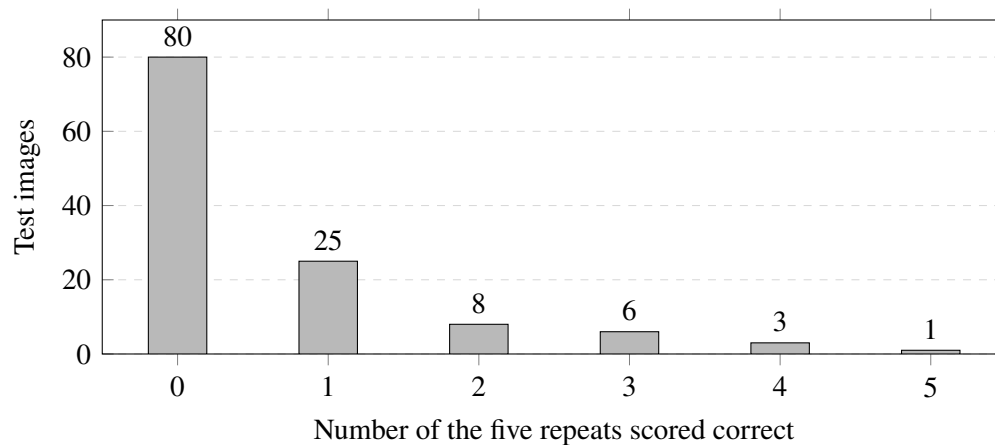


Fig. 3. System C repeat consistency across 123 test images. On 80 images no repeat succeeded and on only one image did all five, leaving 42 images where the same model on the same pixels sometimes passed and sometimes did not.

Self-agreement carries usable signal. Restricting to the 21.1% of images where at least 40% of repeat pairs agreed lifts best-of-five accuracy to 57.7% on those images, against 35.0% at full coverage. The match with System A's single-prediction 57.7% is a coincidence. This describes which of several repeated calls to trust, not a fourth accuracy figure to set against Systems A and B.

System D

System D chose a passing candidate on 18.7% of 615 calls (per repeat 17.1, 17.1, 17.9, 20.3, and 21.1%), with a best-of-five of 39.0%. Every called reply parsed (560 of 560; the other 55 calls belong to the eleven images with no candidates and count as misses). Table 6 sets that against the floor and ceiling of the same menu, and Fig. 4 shows one test image with the marks the model saw.

Table 6. System D against the floor and ceiling of its own candidate menu, and against System C, on the same 123 test images. Intervals are object-clustered.

	Accuracy	95% CI
System D, one call per image	18.7%	[9.5, 29.7]
Random choice among candidates	20.1%	[14.6, 26.6]
Any candidate passes (ceiling)	79.7%	[67.2, 90.0]
Geometric rule alone (candidate 0)	55.3%	[41.8, 68.0]
System C, one call per image	12.4%	[8.0, 17.4]

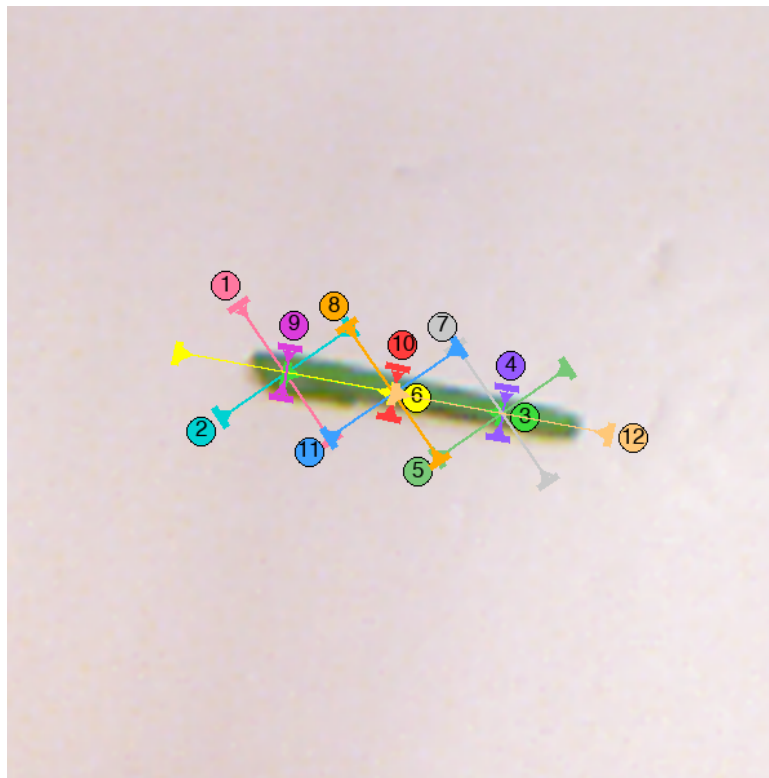


Fig. 4. The marked image for one test object as System D saw it. The model chose mark 6, the yellow end pinch along the bean, on all five repeats, describing it as gripping “the central narrow part of the bean”. Mark 10 crosses the bean at its centre and passes the metric.

Self-agreement across the five repeats, measured on the physical candidate with the numbering re-shuffled each call, was 44.3%, twice System C’s. The preference it agreed on was for the candidate whose jaws travel along the object’s long axis near one end: 342 of 560 choices for that

single candidate, 391 for that orientation at any position. That candidate type passes on 17.0% of test images. The centroid candidate closing across the short axis passes on 60.7% and was chosen 34 times.

The two follow-up menus are in Table 7. On the four-candidate menu drawn at real size the model scored 26.5% against a 24.6% random floor; with every mark the same size, 22.3% against the same floor. Neither paired difference excludes zero (+1.9 [−2.2, +6.3] and −2.3 [−6.3, +1.7]), nor does the difference between the two menus (+4.2 [−0.5, +9.4]). The sparse menu’s 7.8 point lead over the full menu excludes zero, but its floor is 4.5 points higher, so the gain belongs to the menu rather than the model.

Table 7. System D on three candidate menus, same model, same 123 test images, five repeats each. “Long axis” is the share of parsed calls choosing the orientation whose jaws travel along the object’s long axis; chance is 25% on the four-candidate menus.

Menu	Marks	Accuracy	Random floor	Ceiling	Long axis chosen
Full, three positions	12	18.7%	20.1%	79.7%	70%
Centroid only, real size	4	26.5%	24.6%	65.9%	41%
Centroid only, uniform size	4	22.3%	24.6%	65.9%	45%

Cross-system comparison

System A against System B is significant under McNemar’s exact test ($p = 1.4\text{e-}4$), the 38-versus-11 split described below. System C was tested separately against each of its five runs rather than pooled, and the least significant of the five is reported, $p = 5\text{e-}12$ against System A and $p = 1.6\text{e-}20$ against System B.

Every interval quoted so far treats images, or in System C’s case individual calls, as independent. They are not, because the 123 test images sit inside 35 objects. Resampling whole objects with replacement widens all of them. System A becomes [46.2, 68.2], System B [70.8, 87.1], System C [8.0, 17.4], and System C’s best-of-five [24.6, 45.5]. Checking whether two such intervals overlap is a weak test, so the differences were bootstrapped directly on the same resamples. System B leads System A by 22.0 points [9.7, 35.1], System A leads System C’s best-of-five by 22.8 points

[9.2, 35.6], and System A leads System C at one call per image by 45.4 points [33.8, 55.6]. None of those intervals contains zero, and an object-clustered permutation test agrees at $p = 0.003$, 0.006 , and $5e-5$. The ordering of the three systems survives the change of sampling unit. The Wilson intervals reported per system above are image-level and should be read as the narrower of the two.

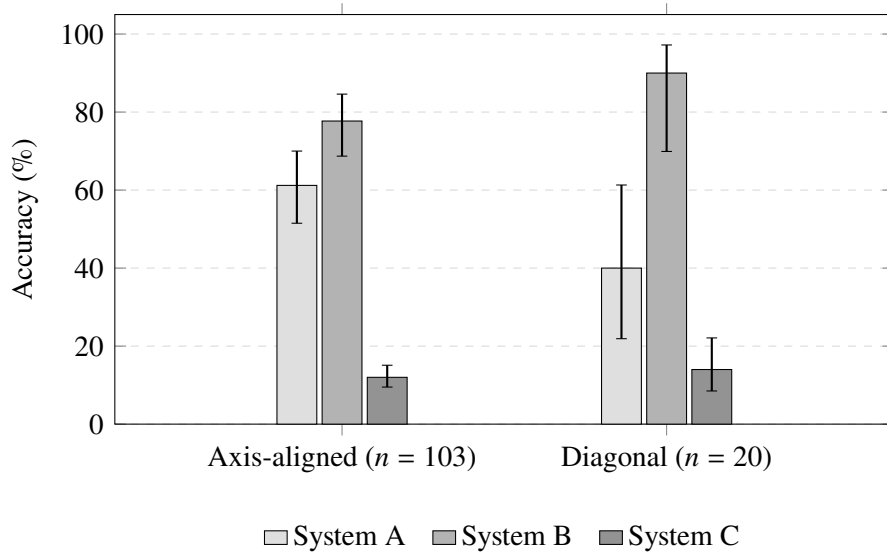


Fig. 5. Accuracy by orientation stratum with 95% Wilson intervals. Axis-aligned, 103 images: System A 61.2 [51.5, 70.0], System B 77.7 [68.7, 84.6], System C 12.0 [9.5, 15.1]. Diagonal, 20 images: System A 40.0 [21.9, 61.3], System B 90.0 [69.9, 97.2], System C 14.0 [8.5, 22.1]. These are image-level rather than object-clustered, so they are the narrower reading. The three systems move in three directions, but the diagonal intervals are already wide enough that only System A's drop, which was predicted from its representation beforehand, should carry weight.

Systems A and B agreed on 74 of 123 images, both correct on 60 and both wrong on 14. Of the 49 they disagreed on, System B alone was correct on 38 and System A alone on 11. Counting System C as correct whenever any of its five repeats passed, all three systems succeeded together on 22 images and failed together on 10. Fig. 6 shows one such image with all three systems overlaid.

All three systems did worst on the images with the most labeled grasps (8 to 25), rather than the fewest, the opposite of what a more-labels-means-easier-match hypothesis would predict. In that group, System A scored 48.0%, System B 64.0%, and System C 11.2%.

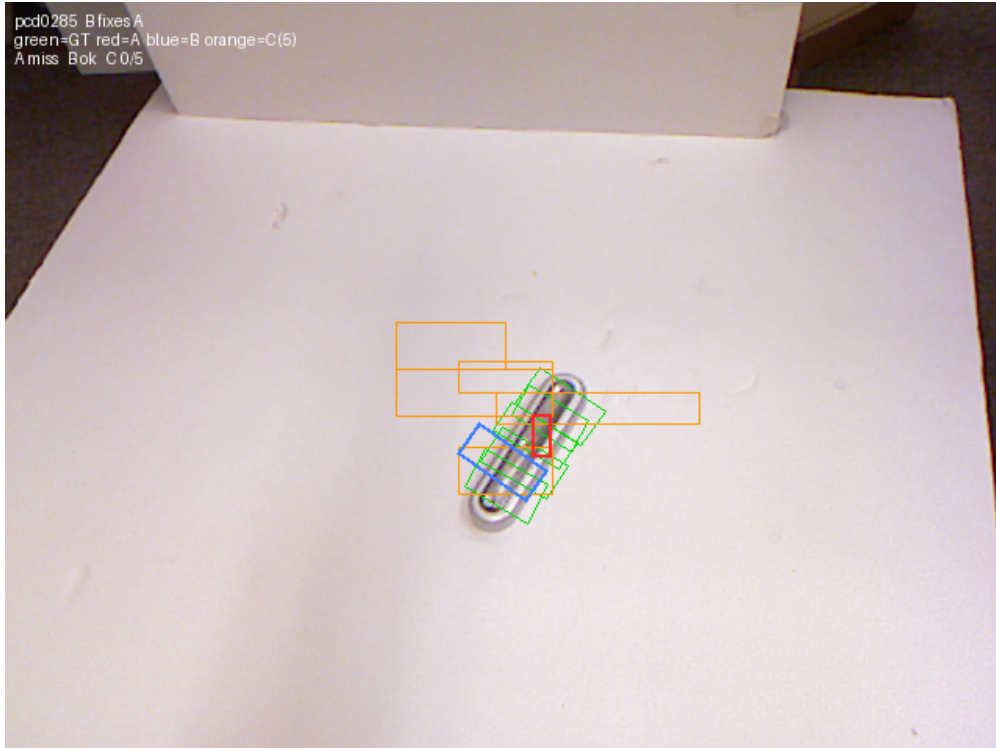


Fig. 6. All three systems on one image. Labeled ground-truth grasps are green, System A red, System B blue, and System C's five repeats orange.

Conclusion

System A's accuracy of 57.7% is lower than System B's 79.7% (84.0% averaged over three seeds under a stable recipe), and higher than System C's 12.4% for zero-shot GPT-4o.

System A places a rectangle on the object often enough to score, though its detector fires on only 49 of 123 images and a geometric fallback carries the rest, and it cannot rotate the gripper at all, since an axis-aligned box has no way to express a diagonal grasp. System B rotates well, to within about eight degrees of the matched grasp on the predictions it gets right, with most of its remaining error coming from placement rather than orientation. System C produced the most distinctive error pattern in this comparison. Its coordinates are frequently not bound to the object its own text has just described correctly, so it fails before either of the other two questions is even reached.

The third failure looked, from System C alone, like a failure to turn a correct visual choice into coordinates. System D tested that reading by taking the coordinates away. Offered a menu

of candidate grasps with a 79.7% ceiling, the same model chose no better than chance, and what it chose was consistent: an end pinch along the object's long axis, which from above resembles a grip on a narrow part and in three dimensions is not one. The error persisted without the coordinate channel and with every mark drawn the same size, so the coordinate output alone does not explain it. The evidence points to the grasp choice, though whether the model reads the marks as intended was not separately measured. The same menu also produced a label-free geometric rule that matches the detector-based baseline while handling diagonal objects, which is the cheapest system in this paper to build. What remains untested is whether depth input, or a model with a stronger three-dimensional prior, changes the choice.

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